Experiment 7

Integration of Advanced Analytics with BI Reports using Microsoft Power BI

Aim: To integrate advanced analytics with Business Intelligence reports using Microsoft Power BI and generate business insights, strategic recommendations, and executive presentations that support organizational decision-making

OBJECTIVES:

1. To analyze existing Business Intelligence reports and dashboards using Power BI.
2. To identify business strengths, weaknesses, and opportunities through data analysis.
3. To generate meaningful business insights using KPIs, charts, and analytical reports.
4. To prepare a Business Insight Report with findings and recommendations.
5. To communicate analytical findings through an executive presentation.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. Power Query Editor
5. DAX (Data Analysis Expressions)
6. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

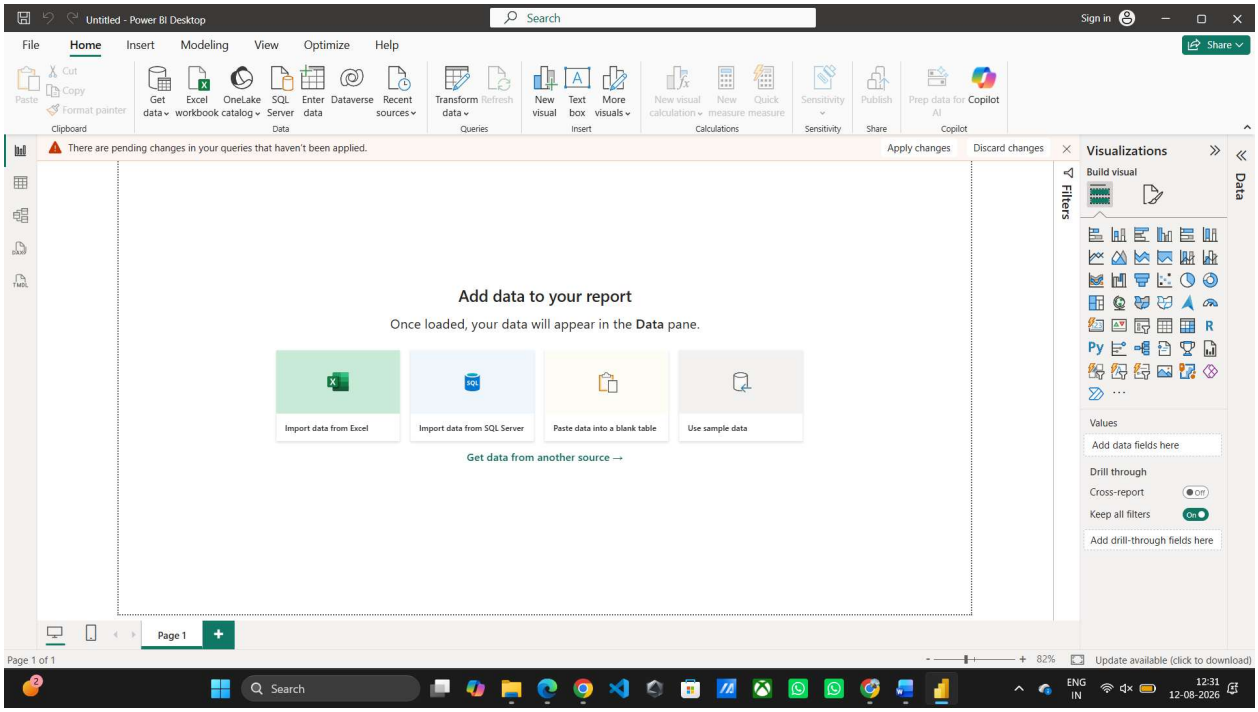
Advanced Analytics in Business Intelligence focuses on transforming data into actionable insights. While traditional BI reports present information, advanced analytics helps identify trends, opportunities, business challenges, and improvement areas. Organizations use analytical reports and dashboards to support strategic planning and decision-making.

Power BI enables advanced analytics through KPI analysis, trend analysis, comparative analysis, ranking, segmentation, and interactive reporting. The ultimate objective is to convert analytical findings into meaningful business recommendations.

PROCEDURE:

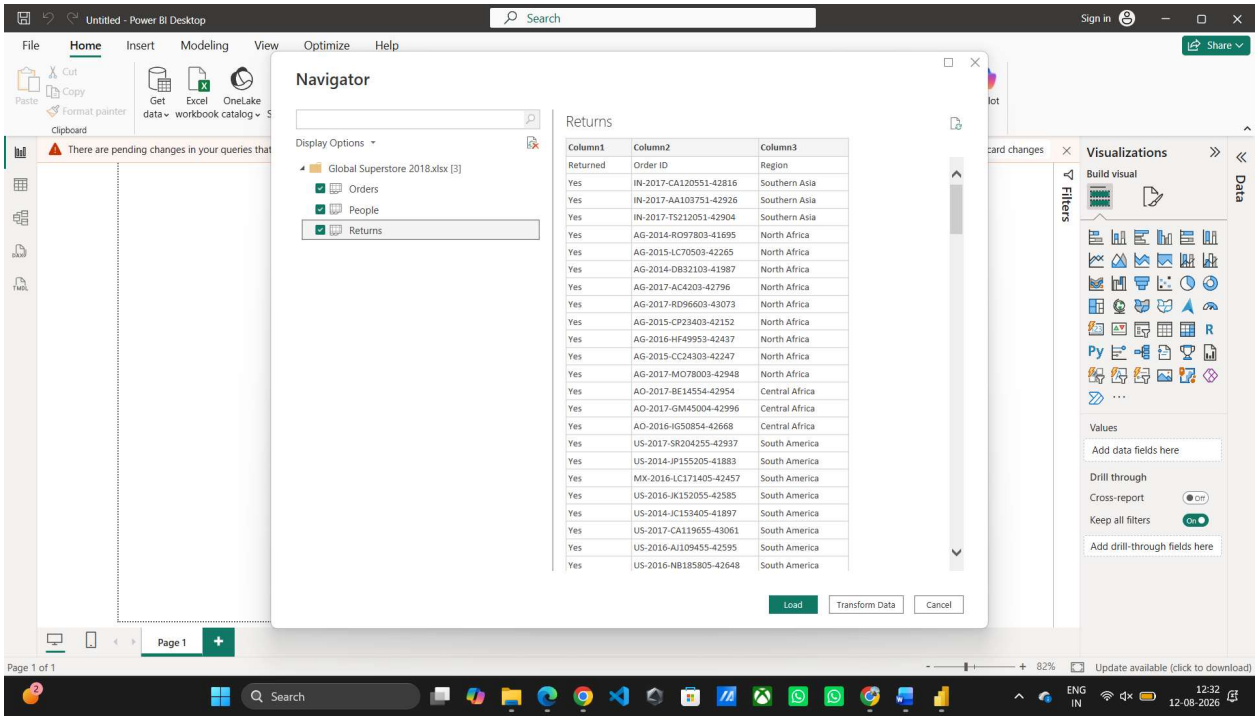
Part A: Dashboard Planning

Step 1: Open the Power BI project created in previous experiments.



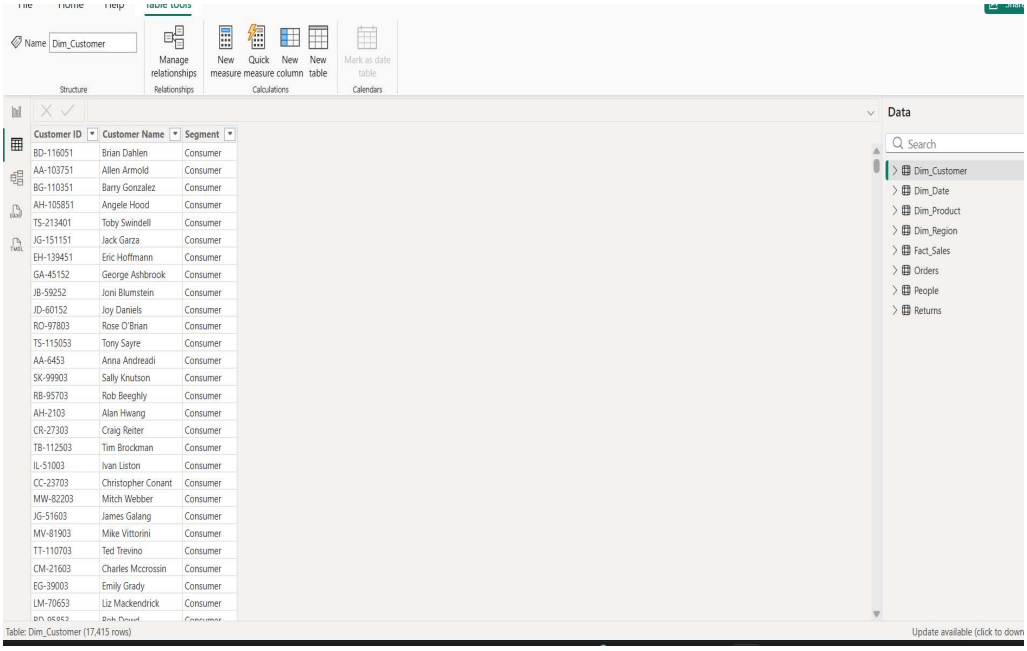
Step 2: Verify the imported tables.

Check the Data/Fields pane and verify that the existing tables, such as `Orders`, `Fact_Sales`, `Dim_Product`, `Dim_Region`, and `Dim_Date`, are available.



Step 3: Create a Table Visual.

1. Open Report View.
2. Select an empty area of the report canvas.
3. Choose the Table visual from the Visualizations pane.



Step 4: Add the required fields.

Add the following fields to the Table visual:

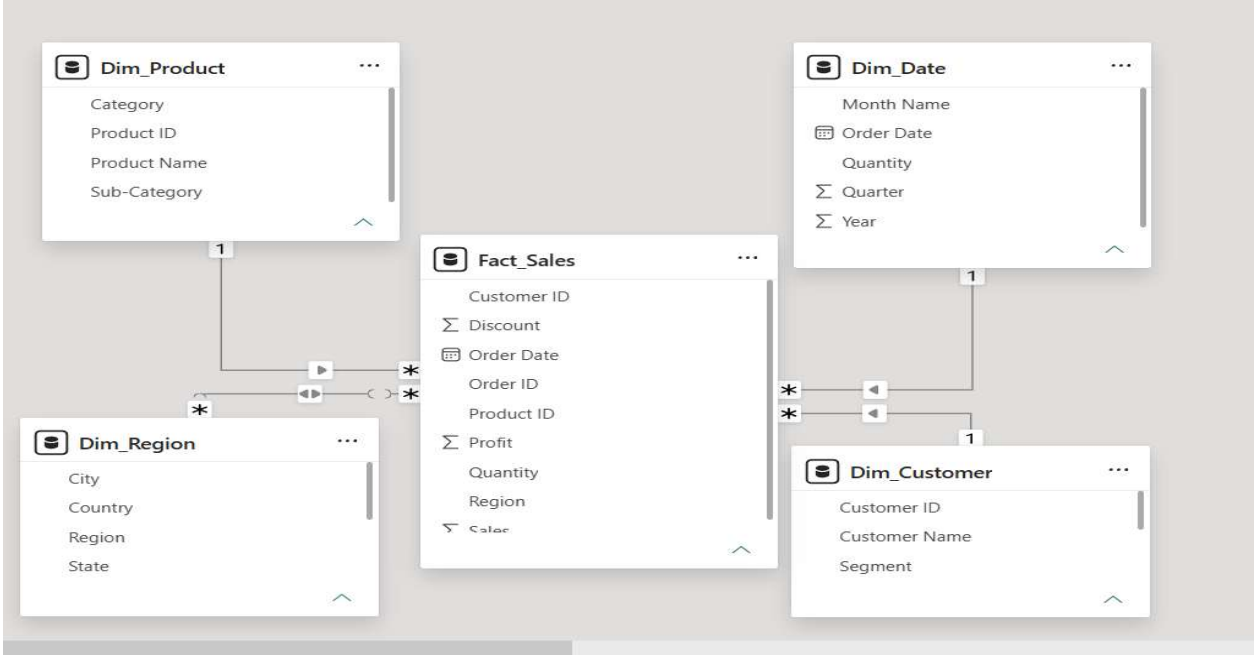
- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit

Select the appropriate fields from the existing tables in the model.

Step 5: Identify key business metrics required by management.

Step 6: Select suitable visualizations for presenting performance indicators.

Step 7: Sketch a dashboard layout before implementation.



Part B: Business Problem Identification

Step 8: Identify Business Strengths

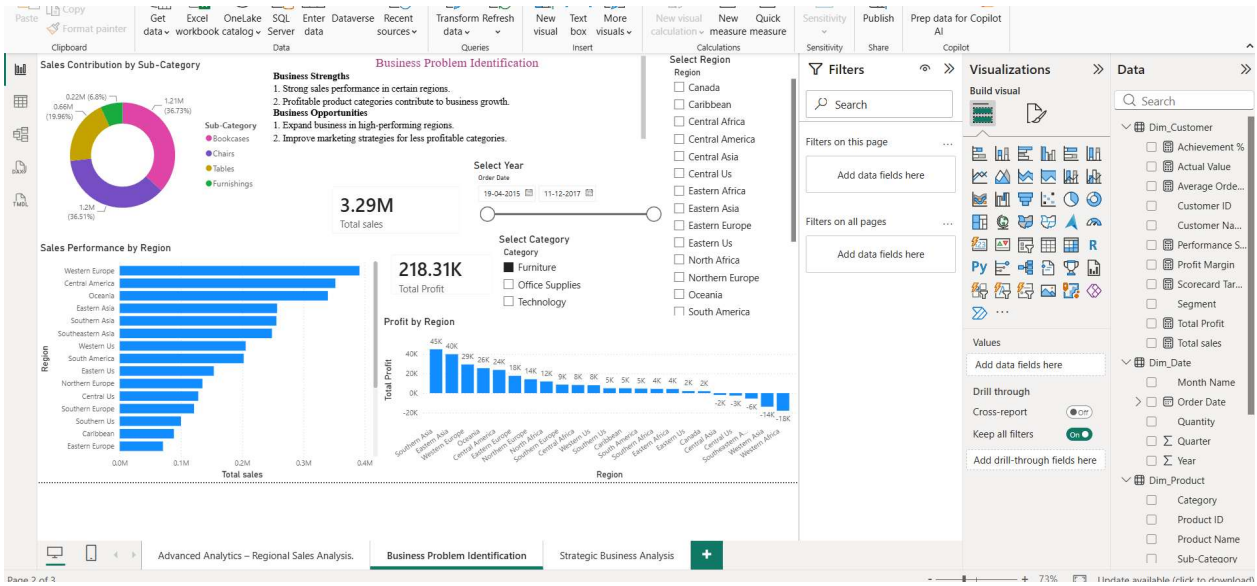
1. Examine the existing Power BI reports and dashboards.
2. Identify areas where the organization performs well.
3. Analyze regions with strong sales performance.
4. Identify product categories generating higher profits.
5. Identify customer segments contributing positively to business performance.
6. Record at least two business strengths based on analytical findings.

Step 9: Identify Business Weaknesses

1. Analyze regions showing low sales performance.
2. Identify product categories generating low profits.
3. Examine customer segments with lower profit contributions.
4. Identify areas where business performance needs improvement.
5. Record at least two business weaknesses based on the available data.

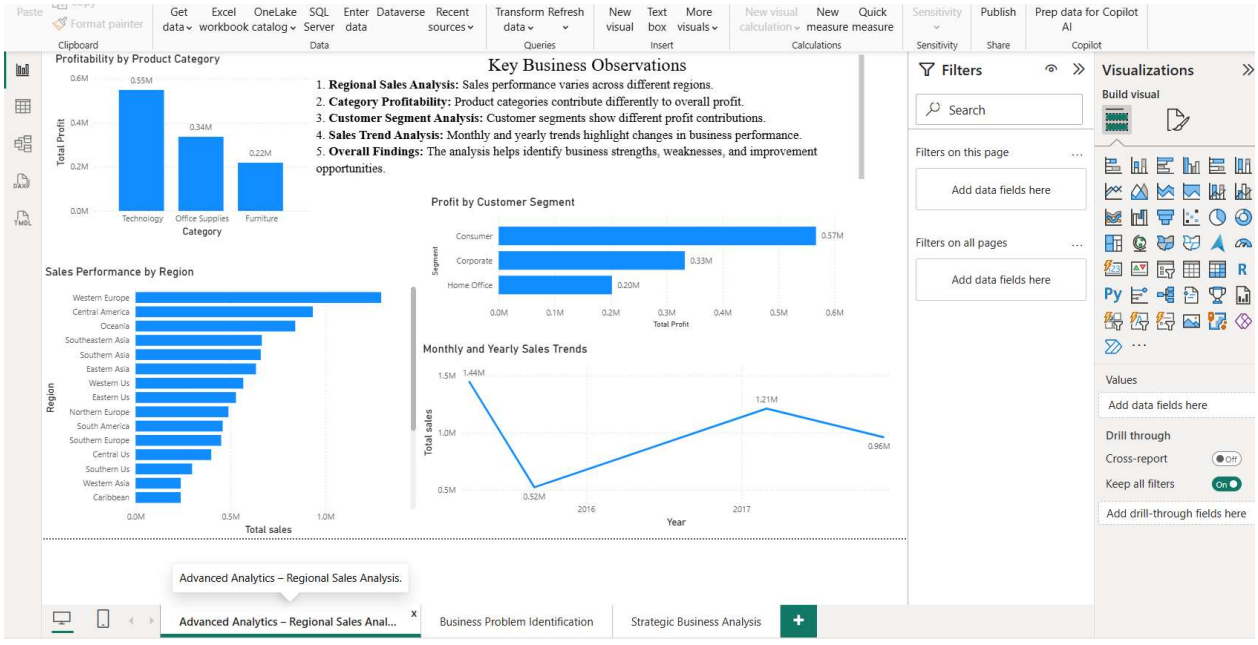
Step 10: Identify Business Opportunities

1. Identify regions with potential for business expansion.
2. Examine opportunities to improve product profitability.
3. Identify customer segments that can be targeted for better performance.
4. Suggest areas where sales and marketing strategies can be improved.
5. Record at least two business opportunities.



Step 11: Support Findings Using Analytical Evidence

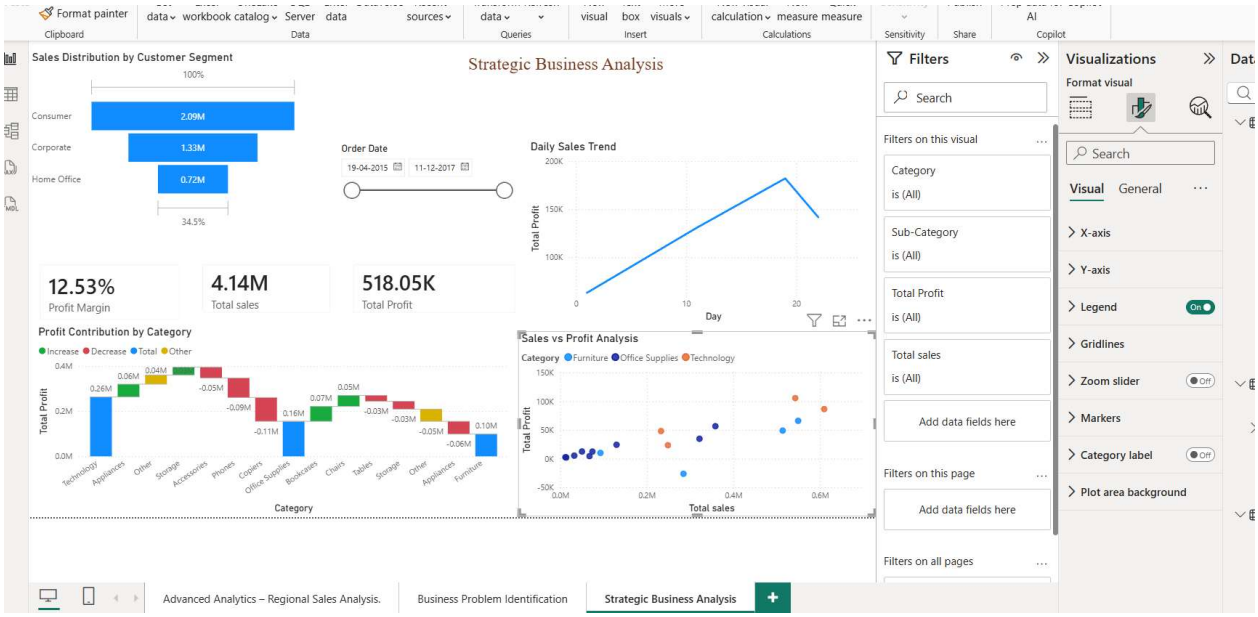
1. Use Power BI charts to support the identified strengths and weaknesses.
2. Use KPI cards to compare important business metrics.
3. Analyze sales and profit values across different regions and categories.
4. Use interactive slicers and filters to examine specific business performance.
5. Present findings using relevant charts, KPIs, and analytical evidence.



Part C: Business Insight Report Preparation

Step 12: Prepare a Business Insight Report

1. Open the Strategic Business Analysis page in Power BI.
2. Analyze the KPI cards showing Total Sales, Total Profit, and Profit Margin.
3. Examine the Monthly Profit Trend line chart to identify changes in profitability.
4. Analyze the Sales vs Profit scatter chart to compare sales and profit across sub-categories.
5. Examine the Profit Contribution by Category waterfall chart.
6. Analyze the Sales Distribution by Customer Segment funnel chart.
7. Use the Region, Category, and Year slicers to explore different business performances.
8. Record the key findings, business strengths, weaknesses, and improvement opportunities.
9. Prepare suitable recommendations based on the analysis.



Result

Advanced analytics techniques were successfully integrated with Business Intelligence reports and dashboards. Business strengths, weaknesses, opportunities, and recommendations were identified and communicated through a professional report and executive presentation.

LEARNING OUTCOMES:

- Apply advanced analytical techniques to business reports and dashboards.
- Analyze organizational performance using BI tools and KPIs.
- Generate business insights from analytical findings.
- Develop strategic recommendations based on data-driven evidence.
- Communicate analytical findings through professional reports and presentations.